The Second Loop: Durable Correction of Memorized Errors in Frozen Language Models

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Abstract

A frozen language model that has memorized a wrong fact repeats it even when the correct fact is placed in its context. We study whether such an error can be corrected *durably* — surviving a full process reload — without touching a single weight, and where the criterion of truth for such a correction must live. We introduce the **scar-survival rate**: the fraction of trap facts on which an external correction mechanism keeps beating the model's memorized instinct across N full reloads, scored by a separate semantic judge over the generated answer. On a frozen Qwen2.5-3B-Instruct subject, a per-fact notebook with contrastive decoding turns 0/14 correct answers into 14/14 (the plain notebook without contrastive decoding also reaches 14/14, so contrastive decoding shows no advantage on accuracy; its effect appears in the probability margin on the correct answer) and holds them across ten reloads — 100% under a deliberately deterministic setup (frozen weights, static notebook, greedy decoding) — and under stochastic-decoding stress (700 samples at temperature 0.7), with 92.9% under up to +500 entries of notebook ballast. The mechanism has one sharp failure mode: it is *blind to truth*. Two to three plausible distractors (two on 13 traps, three on one) placed next to the truth halve survival to 50%. A gatekeeper built from a same-family model cannot close the gap — it inherits the subject's blind spots and reaches 64.3%, equal in this configuration to its own accuracy — whereas a gatekeeper grounded in live Wikipedia plus targeted retrieval engineering restores 92.9% (13/14), and 100% once one genuinely ambiguous trap is disambiguated. The metric ports across three models and two fact domains (100% in all six cells, 540 answers) and discriminates a real correction from cosmetic and placebo variants (100% / 0% / 14.3%). A separate probe asks the opposite question — how rarely external truth can arrive before a self-calibrating frozen forecaster drifts — over 1,646 resolved ForecastBench questions and eleven reveal schedules: any finite contact holds; only zero contact fails. We read the whole arc through Tarski's undefinability of truth as an organizing analogy: in our setting neither the subject nor its same-family copy supplied a working criterion of truth; a properly integrated external source did.

1 Introduction

Instruction-tuned language models carry *memorized instincts*: firmly encoded answers that the model emits with high confidence. When such an instinct is wrong — the tallest mountain measured

*An earlier version is archived at Zenodo, doi:10.5281/zenodo.21845022.

Code and data for all six experiments: scar-survival (Part 1), external-grounding (Part 2), thin-channel (Part 3). Interactive result viewers: huggingface.co/spaces/Laborator/second-loop.

base-to-peak, the first animals sent to space, the exact wording of a famous film line — the model repeats the error even when the correct fact is available to it in context. The standard remedy, weight editing or fine-tuning, changes the object we are trying to study and raises its own durability questions.

This paper asks a narrower and, we argue, more fundamental question. Suppose the model is a *frozen subject*: its weights never change. Can we correct a memorized error with an external mechanism, and will that correction be *durable*: specifically, will it survive a full process reload of the model, or is it only a cosmetic effect that vanishes with process memory? We call a durable correction a **scar**, and we call the whole research program the **Second Loop**: correction that closes back on itself and holds across reloads.

We make the following contributions.

1. **A metric.** The scar-survival rate (Section 3): the fraction of working traps on which the mechanism keeps beating the instinct across N full reloads, scored by a separate semantic judge over the actually generated answer.
2. **A mechanism and its failure mode** (Section 4, Experiment 1). A per-fact notebook plus contrastive decoding lifts a frozen 3B model from 0/14 to 14/14 and holds 100% across ten reloads, but collapses to 50% when the notebook is polluted with two to three plausible distractors (two on 13 traps, three on one). The mechanism is blind to truth.
3. **Where the truth criterion must live** (Section 5, Experiments 2–3). A gatekeeper made from a same-family model reaches 64.3%: it shares the subject's blind spots. Replacing it with a real external source (live Wikipedia) plus targeted retrieval engineering restores 92.9%, and 100% after disambiguating one trap. We connect this to Tarski's undefinability of truth.
4. **Metric validation** (Section 6, Experiments 4–5). The metric ports across three models and two fact domains (100% in all six cells) and discriminates a real scar from cosmetic and placebo controls (100% / 0% / 14.3%).
5. **The opposite bound** (Section 7, Experiment 6). Using resolved forecasting questions as a stream of "truth from the future," we sweep how rarely external feedback can arrive before a self-calibrating frozen system drifts. The collapse is binary: any finite contact holds; only zero contact fails.

A note on honesty of claims, stated up front because it governs how every number below should be read. Several headline figures are 100%. Under a frozen model, a static notebook on disk, and greedy decoding, a full process reload deterministically reproduces the same state, so 100% survival across reloads is partly a property of that determinism. The contribution lies elsewhere: (i) that the protocol is portable across models and domains (Section 6), (ii) that it discriminates real from fake interventions (Section 6), and (iii) that it exposes and then closes a real failure mode — memory pollution — with an external source of truth (Section 5).

2 Related Work

Knowledge editing. Methods such as ROME [12] and MEMIT [13] edit facts by modifying weights. Our setting is deliberately complementary: the weights are frozen and the correction lives entirely outside the model, which lets us ask the durability question without confounding it with weight-update stability. Beyond single-edit stability, sequential editing degrades the edited model [5] and edits ripple unpredictably into related facts [4], an added motivation for keeping corrections outside the weights entirely.

Knowledge conflicts and context faithfulness. When an in-context fact contradicts a model's parametric memory, models frequently revert to the memorized answer. This context–memory conflict is documented from entity-substitution studies [11] through controlled behavioral analyses [20], and is systematized in the survey of Xu et al. [21]. Our traps are a deliberately adversarial instance: fourteen facts on which the subject's parametric prior confidently beats a correct in-context fact. The failure mode of §4 (the corrector amplifying a plausible distractor) is the inter-context variant of the same conflict, arising when the notebook itself carries contradictory entries.

Contrastive and context-faithful decoding. Contrastive decoding was introduced as a generation-quality objective [10] and later directed at factuality, either by contrasting layers [3] or, closest to us, by contrasting a run with and without the context, as in context-aware decoding [16]; CK-PLUG [2] and DeCK [1] refine this contrast for knowledge editing. Our mechanism is a direct implementation of this family; our contribution is the durability protocol around the decoder: reload survival, stress loads, and discrimination controls. Our Confidence Gain indicator (Section 4) follows the knowledge-consistency signal of CK-PLUG [2].

Memory corruption in retrieval-augmented systems. Our counterfeit-notebook stress test is a manual instance of corpus poisoning: a handful of injected passages reliably steers retrieval-augmented answers [22, 23], and certified defenses respond by aggregating over isolated passages [19]. Two differences in aim: our injected decoys are not optimized (two or three hand-written decoys suffice to halve survival), and our defense is scored end-to-end by scar survival across reloads rather than by retrieval metrics. The identity of §5 — protected survival equals the verifier's accuracy — holds exactly in our setting; with $n = 14$ we read this as a hypothesis about the mechanism, not a law. Broader treatments of these risks place corpus poisoning within a wider trustworthiness agenda for retrieval-augmented systems [14]. Closest in spirit to our gatekeeper, Si et al. [17] pair semantic filtering of the corpus with a conflict-aware stage that screens query, candidate answer and retrieved evidence for contradiction before generation, a design that independently arrives at the same decomposition as our conflict detector plus arbiter, though scored by generation robustness rather than by survival across reloads.

Limits of self-correction. Intrinsic self-correction — a model revising its own output without new information — largely fails on reasoning tasks [6], and surveys locate reliable correction in external feedback [15, 7]. Our clone-arbiter result is the fact-level, mechanistic version of that conclusion: a same-family verifier inherits the subject's blind spots and matched its own accuracy in our setting (§5), while an external source restores nearly full survival.

Retrieval-augmented grounding. Part 2 grounds a gatekeeper in live Wikipedia retrieval [9]. Our finding there is specifically about the *precision/recall operating point* of the retriever as the binding constraint, not retrieval in general.

Calibration and forecasting. Part 3 uses ForecastBench [8] as a stream of resolved binary questions and applies per-topic Beta calibration.

3 The Scar-Survival Rate

Let T be a set of *working traps*: questions on which the frozen model M, run in its clean regime (no correction mechanism), confidently produces the wrong answer. Let Mem be an external memory of

correct facts, each tagged as "truth that overrides the memorized answer"; let Mech be a correction mechanism running on top of M (contrastive decoding, retrieval augmentation, ...); let N be the number of full process reloads; and let J be a separate semantic judge that labels each answer `correct` or `incorrect` against the reference fact. For each trap t and reload n, $\text{answer}(t, n) = M(\text{Mech}, \text{Mem})(t)$ and $\text{verdict}(t, n) = J(\text{answer}(t, n), \text{gold}(t))$. Then

$$\text{scar-survival rate} \;=\; \frac{\left|\{\, t \in T : \forall\, n \in \{1, \ldots, N\},\ \text{verdict}(t, n) = \texttt{correct}\,\}\right|}{|T|}. \tag{1}$$

Two design choices matter. First, each reload n is a *separate operating system process*: the model is fully unloaded and reloaded from scratch and the notebook is re-read from disk, so a "cosmetic" correction that lives only in process memory cannot survive. Second, scoring is done by a semantic judge over the actual generated answer, not by keyword matching or by a token-level probability metric; Section 4 shows why the latter is unreliable. Throughout this paper $N = 10$ and J is Qwen2.5-7B-Instruct.

A note on what J does and does not decide. The judge is not asked to establish what is true: it receives the reference fact together with the candidate answer and only checks semantic consistency between them. This is a different task from the arbiter of Section 5, which must label each entry as true or false without being given the answer, and it is why a same-family model is adequate here and inadequate there. In the clean regime the reference handed to J is the same notebook entry the mechanism places in the subject's context, so a survival of 14/14 states that the mechanism carried that entry through to the generated output, not that the answer is correct against an independent source. The stress and control regimes break that coincidence deliberately: under the noisy notebook the context also carries distractor entries while J still scores against the original fact, and in the placebo regime the context carries an irrelevant entry while J scores against the real one. That is what allows those regimes to fall below 100%; the specific values are set by the generated answers themselves (Sections 4 and 6).

4 Experiment 1: The Scar Mechanism

Setup. The frozen subject is Qwen2.5-3B-Instruct, run on a single RTX 3090 Ti. We screened both Qwen2.5-7B-Instruct and Qwen2.5-3B-Instruct over the same 58 candidate traps (myths, geography, quotations, niche facts, "false friends") with the same automatic marker-based classifier, which labels each answer `correct`, `wrong_instinct`, or `ambiguous`. Under that single criterion the two models fire the memorized error at the same rate — 6/58 = 10.3% each — but they differ in how cleanly they get the rest right: the 7B leaves no ambiguous answers, whereas the 3B leaves 7, so 13/58 (22.4%) of its answers are not unambiguously correct. We chose the 3B because it also hallucinates its own wrong answer readily, which is what the mechanism must override. A manual review of all 58 of the 3B's answers (not applied to the 7B) yields the **14 working traps** used throughout this paper. Thirteen come from the fired and ambiguous buckets; one (#46) was promoted from an answer the classifier scored `correct` because the generated text contradicts itself arithmetically: it answers "No," then reports a Venusian day of 243 Earth days against a Venusian year of approximately 225, yet calls the day *shorter* than the year. The external memory is a notebook holding, for each trap, the correct fact.

Correction mechanism. At each decoding step the model is evaluated twice — with the fact in context and without it (pure instinct) — and the next-token distribution is shifted toward the

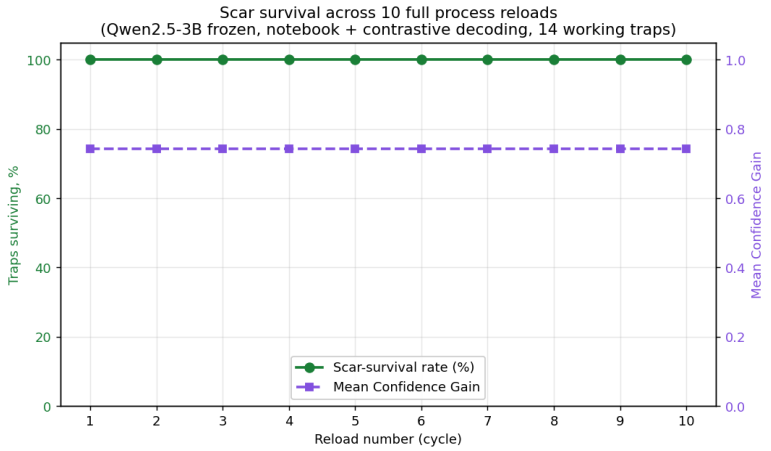


Figure 1: Scar survival across ten full process reloads (Experiment 1): 14/14 traps held in every cycle.

fact:

$$\text{logits} \;=\; (1+\alpha)\cdot \text{logits}_{\text{with fact}} \;-\; \alpha \cdot \text{logits}_{\text{instinct}}, \qquad \alpha = 1.0, \tag{2}$$

with a plausibility threshold $\beta = 0.1$ that prevents amplifying low-probability nonsense. We also report **Confidence Gain** (after CK-PLUG [2]), the increase in probability mass on the correct answer, $P_{\text{contrastive}} - P_{\text{instinct}}$, as an indicator of override strength.

Three modes. Scored by the separate 7B judge over 14 traps: pure instinct gives 0/14 (0%); the plain notebook (fact in context) gives 14/14 (100%); contrastive decoding also gives 14/14 (100%), with a mean Confidence Gain of +0.744. Because the plain notebook already reaches 14/14, contrastive decoding shows no advantage on the accuracy endpoint; its effect appears in the probability margin on the correct answer. On trap #37 ("first animal in space") the probability of the canonical correct answer stays at 0.017 under the plain notebook even though the generated answer is judged correct, whereas contrastive decoding raises it to 0.899.

Scar survival across ten reloads. Running the strict protocol (each cycle a separate process), the scar-survival rate is 100% in every one of the ten cycles (14/14 each time), and the Confidence Gain is bit-for-bit 0.7439 in every cycle (Figure 1). As stated in the introduction, this exactness is a consequence of the deterministic setup; the value of the run is that it establishes the protocol and the baseline against which the stress tests and the discrimination controls (Section 6) are read.

Stress tests. We then probe the override under pressure on the notebook itself (Table 1, Figure 2). It survives a growing notebook (up to +500 ballast entries, 92.9%) and stochastic decoding (temperature 0.7, 700 samples, 100%), and it survives even the stickiest instincts (12 traps with $P_{\text{instinct}} \approx 0$, 100%). It fails under a *noisy notebook*: with two to three plausible distractor entries next to the truth, survival drops to 50% (7/14). Under this load the model copied the false distractor verbatim ("3 seconds," "K2," "Play it again, Sam," "Laika," "Steamboat Willie"), and on two traps the Confidence Gain went negative.

A methodological caveat. Confidence Gain overestimates robustness under noise. It compares the correct answer against a single canonical wrong one, but free generation can latch onto a

Table 1: Experiment 1 stress tests (survival by 7B judge over the generated answer).

Load	Survival	Verdict
Baseline (clean, 10 reloads)	100%	—
1. Growing notebook (+50/100/200/500)	92.9%	holds; Gain stable 0.65–0.72
2. Noisy memory (2–3 distractors)	50% (7/14)	**fails**
3. Stochasticity (temp=0.7, 700 samples)	100%	holds, no spread
4. Sticky instincts ($P_{\text{instinct}} \approx 0$, 12 traps)	100%	holds, Gain +0.58

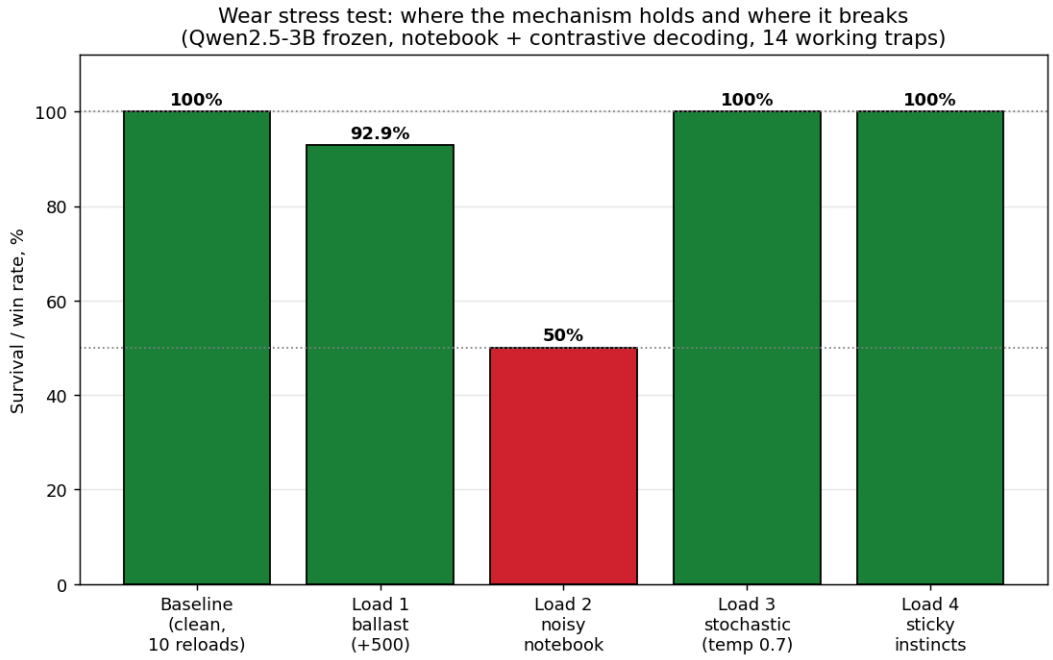


Figure 2: Where the mechanism holds and where it breaks (Experiment 1). Memory pollution, not wear, is the failure mode.

third distractor outside that comparison (on trap #20 the metric read +0.90 while the model generated "K2"). The honest measure is therefore the semantic judge over the actual answer; we keep Confidence Gain only as an indicator of override strength, not as proof of correctness. This is why the metric in Section 3 is defined over J, not over probabilities.

Takeaway. The ceiling of the mechanism is *notebook quality*, not its size, decoding randomness, or instinct strength. Contrastive decoding breaks through the most stubborn instinct, but it will amplify two to three plausible distractors (two on 13 traps, three on one) sitting next to the truth just as readily, especially when a distractor coincides with the model's own instinct. This motivates Part 2.

5 Guarding the Notebook: Where the Truth Criterion Must Live

The failure mode of Section 4 is memory pollution. We place a *gatekeeper* at the entrance to the notebook whose job is to reject false or conflicting entries before they reach the corrector.

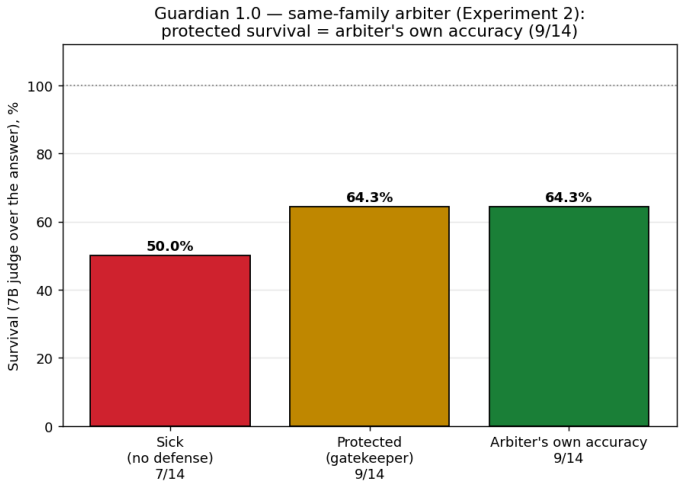


Figure 3: The clone ceiling: survival under Guardian 1.0 (64.3%) exactly equals the fraction of traps on which the arbiter itself kept the correct fact (9/14). In this configuration protected survival equals arbiter accuracy; we read this as a hypothesis about the mechanism, not a general bound.

The gatekeeper has three parts: a conflict detector (pairwise comparison of entries for the same question), an arbiter (which adjudicates flagged conflicts), and a quarantine (rejected entries are isolated from the working notebook). The conflict detector caught the contradiction in every noisy entry (14/14); the interesting variable is the *arbiter*.

Guardian 1.0 — a same-family arbiter (Experiment 2). When the arbiter is a second model of the same family (Qwen2.5-7B judging entries for a Qwen2.5-3B subject), survival rises from the polluted 50% only to 64.3% (9/14). Protected survival (9/14) equals the fraction of traps on which the arbiter itself kept the correct fact (9/14) (Figure 3). We read this as a hypothesis about the mechanism, not a general bound. The arbiter and the semantic judge are the same model (Qwen/Qwen2.5-7B-Instruct). The arbiter failed on precisely the stickiest facts where the subject failed (Mauna Kea; the length of a day on Venus; fruit flies vs. Laika; the misquotes from *Snow White* and *Casablanca*), because a same-family model shares the subject's blind spots. On one trap (#46) the defense even regressed, quarantining a correct fact it judged false. In this configuration the clone did not supply the truth criterion.

Guardian 2.0 — an external source (Experiment 3). We replace the model arbiter with live Wikipedia retrieval and a grounded reader. The survival arc is shown in Table 2 and Figure 4. A raw external source lifts survival from 64.3% to 71.4% by fixing traps on which the clone failed. A blunt "make retrieval stronger" pass does nothing (71.4%→71.4%): it merely trades false admissions for false quarantines: a precision/recall wall. The gain comes instead from three *targeted* fixes: (i) a verbatim-match check for quote facts; (ii) a relevance gate that rejects coincidental namesake articles (a query about a *Snow White* line is no longer "confirmed" by the 2012 film *Mirror Mirror*); and (iii) a soft threshold that keeps a doubtful truth at reduced trust instead of quarantining it, reserving quarantine for entries the source explicitly contradicts. These reach 85.7%. A final calibration of the relevance gate, plus rewriting one genuinely ambiguous trap, reaches 100% (13/14 on the original items). The 2.1–2.3 fixes were motivated by failure cases within the same 14 traps (e.g.

Table 2: Full survival arc under a polluted notebook (Experiments 1–3), scored by the separate judge over the generated answer.

Version	Verifier	Scar survival
Sick (no defense)	—	50.0%
Guardian 1.0	model arbiter (clone)	64.3%
Guardian 2.0	Wikipedia (retrieval)	71.4%
Guardian 2.1	wiki + brute strengthening	71.4%
Guardian 2.2	wiki + three targeted fixes	85.7%
Guardian 2.3	wiki + final calibration	**100.0%**†

† includes disambiguating one trap (#46); without it, 92.9% (13/14).

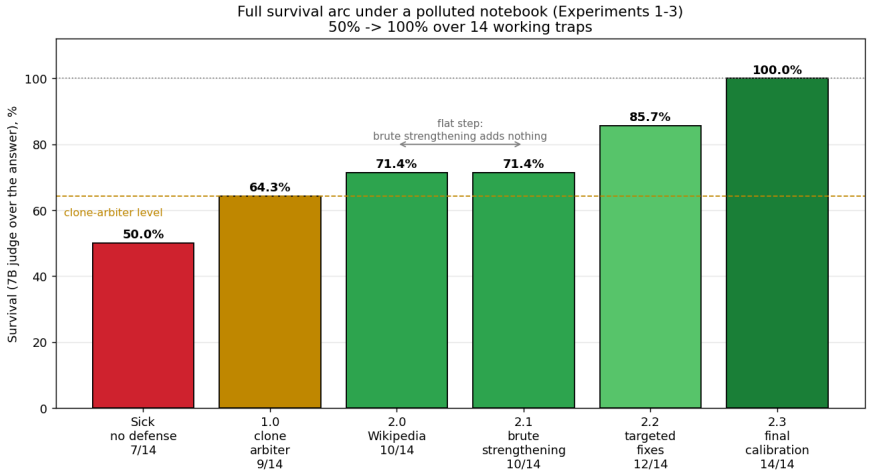


Figure 4: From 50% to 92.9%, and to 100% after disambiguating one trap: healing a polluted notebook with a real external source of truth plus targeted retrieval engineering (Experiment 3).

the *Snow White* namesake article, trap #46); an untouched held-out set is needed to measure how much of the gain generalizes.

Interpretation. Three lessons. (a) The truth-blindness limit is closed by a *combination* — external source *and* retrieval engineering *and* clean, unambiguous traps — not by any single trick; remove one and the ceiling drops. (b) Not every trade-off is fundamental: the apparent precision/recall conflict at 2.1 dissolved once the tool was made smarter (keep by default, drop only explicit namesakes). (c) A real external source audits the *test set*, not just the model: trap #46 ("is a day on Venus longer than its year?") was ambiguous between the sidereal day (243 days, longer than the 225-day year) and the solar day (∼117 days, shorter); the clone never noticed, but the external channel "confirmed" both readings because both are true, forcing us to disambiguate the question. We read this through Tarski's undefinability of truth [18] as an organizing analogy, not a formal claim: a sufficiently expressive system cannot define its own truth predicate, and in our setting neither the subject nor its same-family copy could supply a working criterion of truth; an external source, properly integrated, could.

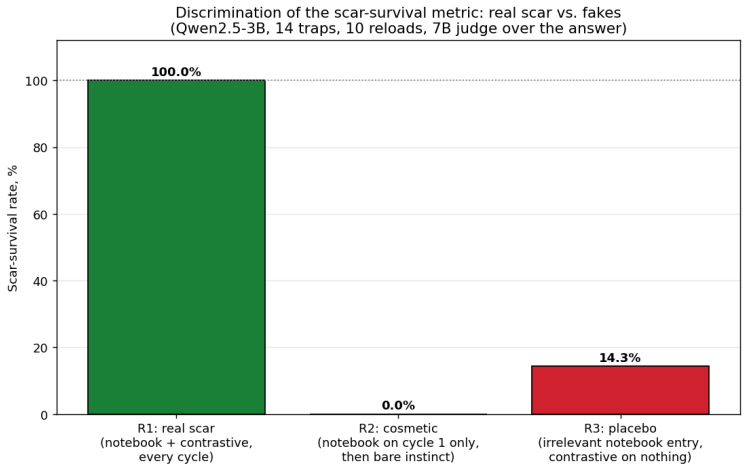


Figure 5: Metric discrimination (Experiment 5): real scar (100%) vs. cosmetic (0%) vs. placebo (14.3%) over ten reloads.

6 Does the Metric Mean Anything? Universality and Discrimination

A metric that reads 100% is only useful if it also (a) ports beyond one model and one fact set and (b) returns *low* scores on interventions that are not real. We test both.

Universality (Experiment 4). We run the full protocol on a 3×2 grid: three instruction-tuned models (Qwen2.5-3B-Instruct, Qwen2.5-7B-Instruct, Phi-3.5-mini-instruct) $\times$ two fact sets (set A: 58 misconceptions; set B: 20 geography/records), with 10 reloads per cell and 7–10 working traps per cell (540 answers total). The scar-survival rate is 100% in every one of the six cells under the same deterministic setup. On a single RTX 3090 Ti the end-to-end run takes $\sim$3–4 hours, dominated by repeated model loads.

Discrimination (Experiment 5). On Qwen2.5-3B-Instruct we run three regimes head-to-head, 10 reloads each (30 cycles, 420 judged answers; Figure 5):

- **R1 — real scar**: notebook reconnected every cycle, contrastive decoding with the refuting fact. Result: **100%**.
- **R2 — cosmetic**: cycle 1 uses the notebook, cycles 2–10 run bare instinct. Result: **collapses to 0%** immediately after cycle 1.
- **R3 — placebo**: notebook reconnected every cycle but every entry is an irrelevant administrative sentence. Result: **14.3%**. R3 differs from the Experiment 1 baseline in system prompt and decoding path. On traps #23 and #36 the placebo-conditioned run rejects the irrelevant note and names the correct answer (Leif Erikson; Valentina Tereshkova), whereas the plain-prompt baseline answers Columbus and Savitskaya. The 14.3% is therefore the placebo condition's own rate, not a chance baseline.

The metric returns high scores only on the real mechanism and low scores on the two fakes, so it tracks the mechanism rather than the model's general competence.

7 The Thin Channel: How Rarely Truth Can Arrive

Parts 1–2 showed that a live external source repairs a corrupted memory on the evaluation traps. Part 3 is a methodologically separate probe of the same wall from the opposite side: instead of asking whether external truth can fix a frozen system, it asks how rarely that truth can arrive before the system drifts. The setup, metric, and object differ deliberately; what carries over is the question of where the criterion of truth must live.

Setup. We use ForecastBench — 1,646 resolved binary forecasting questions (July 2024–December 2025) — as a stream of "truth from the future." A frozen Qwen2.5-3B-Instruct emits a raw $P(\text{YES})$ for each question at its freeze date; the outcome is hidden and revealed only on a schedule. On top of the frozen model sits a per-topic Beta calibration (9 topics, saturation $K = 10$) with no weight updates. We split 1,316 train / 330 hold-out by freeze date and report hold-out Brier score. The same stream is run independently through eleven reveal schedules from daily to "never."

The raw model is a poor forecaster — calibration does the work. On 250 questions the raw 3B has Brier 0.354, worse than an "always NO" baseline (0.340) and much worse than "always topic base rate" (0.224). On the 330-question hold-out, an end-of-stream ablation gives: raw model 0.298; topic base-rate only 0.1890; the Beta mixture 0.1889. This ablation scores every hold-out question with the final Beta, built from all 1,316 train outcomes, including outcomes that resolved after a hold-out question's forecast date; it is therefore an optimistic, look-ahead figure. Under the online daily schedule, where each hold-out forecast sees only outcomes already revealed, the mixture scores 0.2050 and base-rate-only 0.2103. The mixture beats the raw model by 0.109 Brier offline and 0.093 online, but beats base-rate-only by only 0.0001 offline and 0.0054 online: in both regimes nearly all of the usable signal is the topic base rate, and the raw model adds at most 0.0054 Brier on top of it. We state this plainly, without dressing the calibrator up as a competent forecaster.

The collapse is binary. Sweeping the reveal schedule (Table 3, Figure 6), all ten *finite* schedules, from daily down to once a year, land in hold-out Brier $[0.205, 0.220]$. Only "never" collapses to 0.298, bit-for-bit the raw 3B baseline. Neither frequency nor delivered volume is the binding constraint in this regime. The 270-day schedule delivers 220 resolutions and reaches Brier 0.216; the daily schedule delivers 1,316 and reaches 0.205. A six-fold difference in delivered signal moves the score by 0.011, while the gap between any finite schedule and no contact at all is 0.078 to 0.093. With 9 topics at $K = 10$ the calibration saturates after a few hundred resolutions, so the experiment separates contact from no contact and says nothing finer.

8 Limitations and Scope of Claims

- **Determinism inflates the headline numbers.** The 100% survival in Experiments 1 and 4 is partly a consequence of frozen weights, a static notebook, and greedy decoding: a process reload reproduces an identical state. The claims we stand behind are the *portability* (Experiment 4), the *discrimination* (Experiment 5), and the *failure-and-repair* arc (Experiments 1–3), not the 100% as such.
- **Confidence Gain is not proof of correctness**; it overestimates robustness under noise (Section 4). All survival numbers are scored by a semantic judge over the generated answer.

Table 3: Hold-out Brier ($N = 330$) across eleven reveal schedules (Experiment 6). Any finite contact holds; only "never" collapses to the raw baseline.

Schedule	Period (days)	Hold-out Brier	Total revealed
day	1	0.205	1316
3 days	3	0.205	1315
week	7	0.211	1313
2 weeks	14	0.210	1313
month	30	0.210	1277
2 months	60	0.220	1173
quarter	90	0.220	1022
half year	180	0.209	417
270 days	270	0.216	220
year	365	0.220	417
never	—	**0.298**	0

Total revealed is not monotone in the period. Reveals fire on days that are multiples of the period within a fixed 535-day window, so the last effective reveal falls on day 360 for the 180-day schedule, day 270 for the 270-day schedule, and day 365 for the 365-day schedule; resolutions still pending when the window closes are never delivered. Because no question resolves between days 361 and 365, the 180-day and 365-day schedules deliver the same 417 resolutions, while the 270-day schedule delivers only 220. The column reports how much signal reached the calibrator inside the simulated window, not how often the schedule fires. Equal volume does not imply equal score, but the differences among finite schedules should not be read as an ordering. On the 330-question hold-out the spread between them is 0.015 Brier (0.2050 to 0.2202). Per-question outcomes were not retained in the sweep, so we report this spread descriptively and do not test pairwise differences. Where a mechanism is identifiable it is one of timing rather than volume: the 180-day schedule delivers its first batch before the hold-out period begins, whereas the 365-day schedule delivers nothing until day 365, leaving its earliest hold-out forecasts uncalibrated (64 of the 330 hold-out forecasts fall before day 365). The claim we stand behind is the gap between any finite schedule and no contact at all, which is 0.078 to 0.093, about five times the spread among finite schedules.

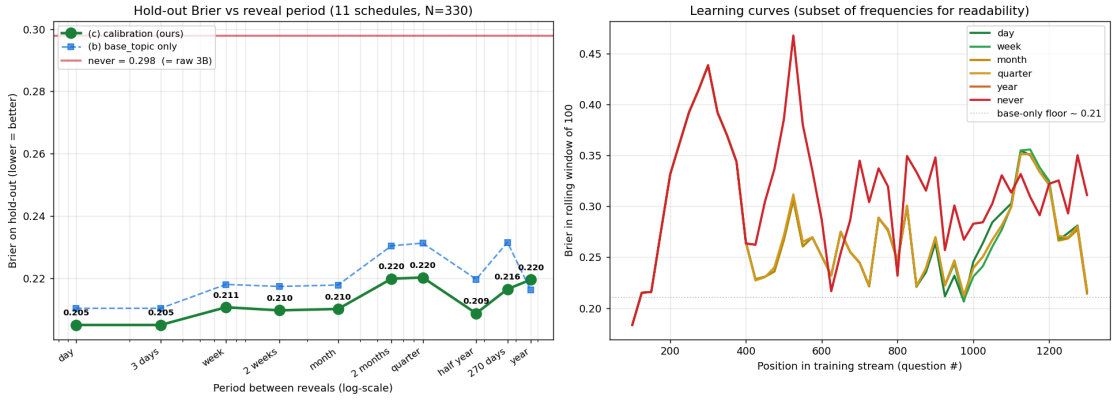


Figure 6: The thin channel (Experiment 6): hold-out Brier vs. reveal period. Every finite schedule holds near 0.21; only zero contact collapses to the raw baseline 0.298.

- **Scale.** Experiments center on a 3B subject with 14 working traps; Experiment 4 extends to 7B and Phi-3.5-mini and to a second fact set, but the regime is small-model, small-trap-set by design (the effect is only visible where the model actually errs).
- **Judge dependence.** The judge is Qwen2.5-7B-Instruct, the same family as the subject. This is the same family whose arbiter we show to be inadequate in Section 5, and the distinction matters: the arbiter must supply a truth criterion, whereas the judge is given the reference fact and only checks consistency against it (Section 3). For the knowledge-editing traps this is adequate, but the judge is itself a model, and the reference it checks against is authored by us rather than drawn from an independent source; a human-audited subset would strengthen the claim. Note that the external source of Section 5 does not change this: the gatekeeper filters what enters the notebook, whereas the reference J scores against remains the entry itself. The judge and the Guardian 1.0 arbiter use the same model (Qwen/Qwen2.5-7B-Instruct, revision not pinned). For trap #46 the reference entry itself was revised (Section 5).
- **Thin channel is a single calibrator.** The binary-cliff result holds for per-topic Beta calibration with 9 topics at $K = 10$; a richer calibrator with more capacity might show a graded, rather than binary, dependence on frequency.
- **Guardian 2.x was tuned on the evaluation traps.** The 2.1–2.3 fixes were motivated by failure cases within the same 14 traps they are scored on; an untouched held-out set is needed to measure generalization.
- **The R3 placebo has a confound.** R3 changes the context content together with the system prompt and the decoder (contrastive instead of plain greedy), so its rate cannot be attributed to the placebo note alone.
- **The thin-channel offline ablation uses look-ahead outcomes.** The end-of-stream figures (0.1889 / 0.1890) use train outcomes that resolved after some hold-out forecast dates; the online daily-schedule figures (0.2050 / 0.2103) do not.
- **Statistical resolution.** With 14 traps every percentage moves in steps of 7.1%; a 14/14 result carries an exact 95% Clopper–Pearson lower bound of roughly 77%. All headline percentages should be read at that resolution.

9 Conclusion

A memorized error in a frozen language model can be corrected durably, without touching a weight, by an external notebook (with or without contrastive decoding), but the mechanism is blind to truth, and two to three plausible distractors (two on 13 traps, three on one) next to the truth halve its survival. In our setting neither the model nor a same-family copy supplied the missing truth criterion; a real external source did, provided its retrieval operating point is engineered rather than merely "strengthened." The scar-survival rate metric ports across models and domains and discriminates real corrections from cosmetic and placebo ones. And when external truth is the healing agent, the low-capacity calibrator used here saturates after a few hundred resolutions: within that regime neither arrival rate nor delivered volume separates the schedules, and only zero contact fails. Across the three parts the pattern is the same: the correction held only when its criterion came from outside the model.

Reproducibility

Every number above is a verbatim output of, or arithmetic on, a committed run; derived numbers are recomputed by `derived_numbers.py`. Part 1 (Experiments 1, 4, 5): `https://github.com/SergheiBrinza/scar-survival`. Part 2 (Experiments 2, 3): `https://github.com/SergheiBrinza/external-grounding`. Part 3 (Experiment 6): `https://github.com/SergheiBrinza/thin-channel`. Subject model Qwen2.5-3B-Instruct (frozen); separate judge Qwen2.5-7B-Instruct. ForecastBench data is not redistributed here; see DATA.md in the thin-channel repository for the download procedure. ForecastBench is CC BY-SA 4.0 [8].

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